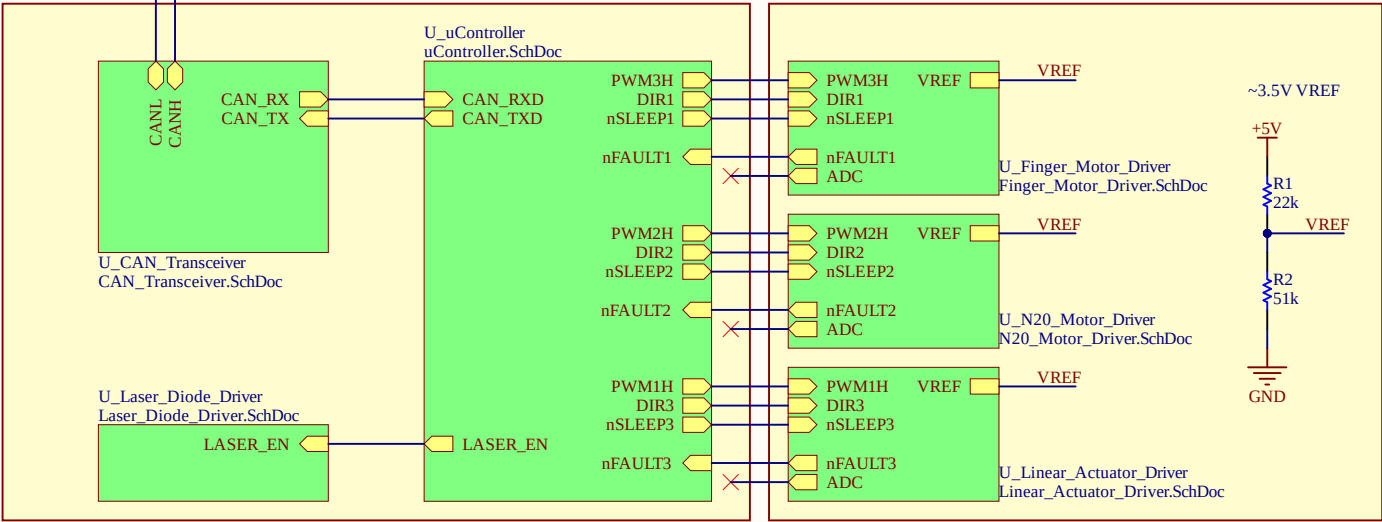
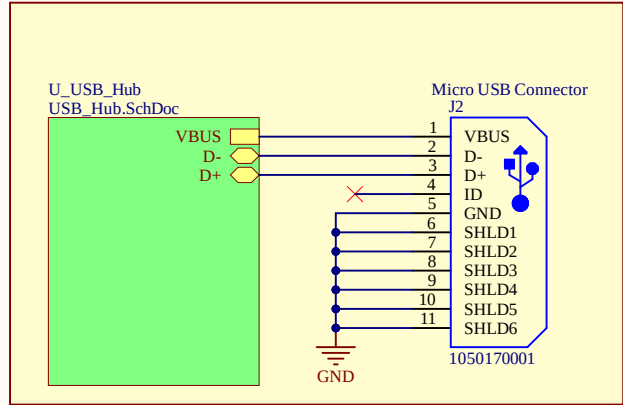
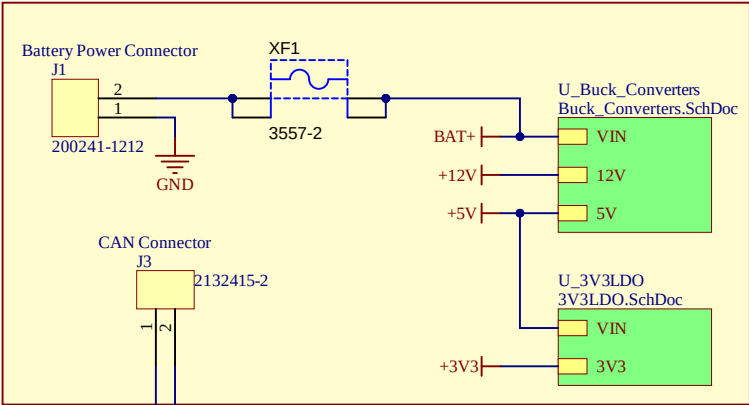
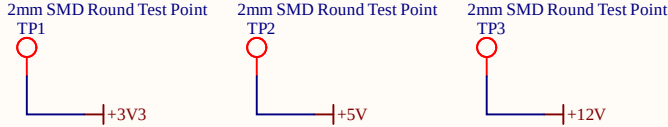
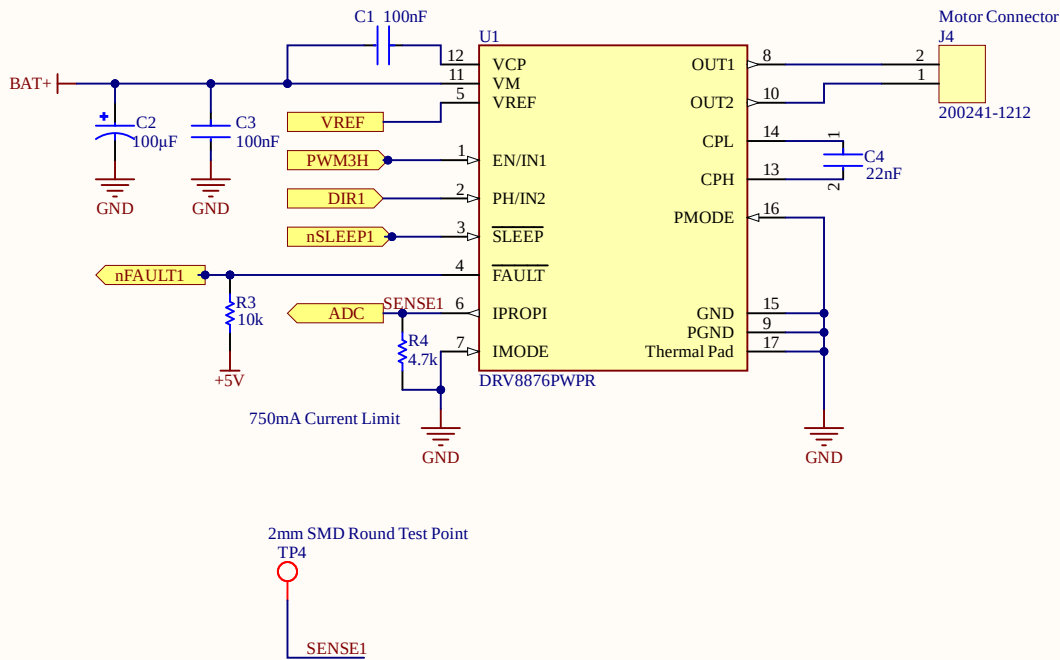




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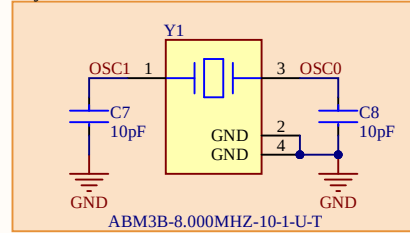
100μF Bulk Capacitor Calculation

$$\frac{1}{2}C_{bulk}\Delta V^2 > \frac{1}{2}LI^2$$
$$V_{motor} = 24V, \Delta V = \frac{20}{100} \times 24 = 4.8, L = 2.3mH, I_{stall} = 3A$$
$$C_{bulk} = \frac{(2.3 \cdot 10^{-3})3^2}{4.8^2} \approx 1 \cdot 10^{-4}F$$



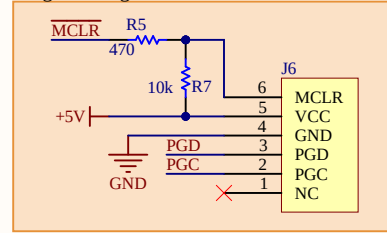
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File:	Finger_Motor_Driver.SchDoc	Drawn By:

Crystal Oscillator

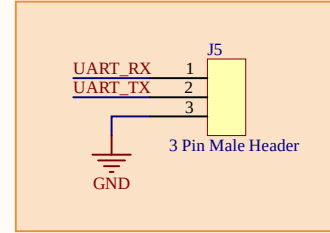


Load Capacitance, $CL = 10pF$
[Assume] $C_{stray} = 5pF$
 $C1 = C2 = 2(CL - C_{stray})$
 $C1 = C2 = 2(10 - 5) = 10pF$
[Caps may require tuning]
Credit to Harrison on CSI serialiser board

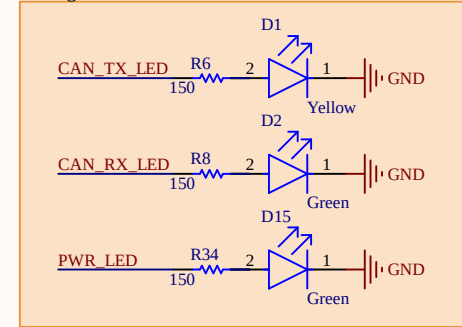
Programming Connector



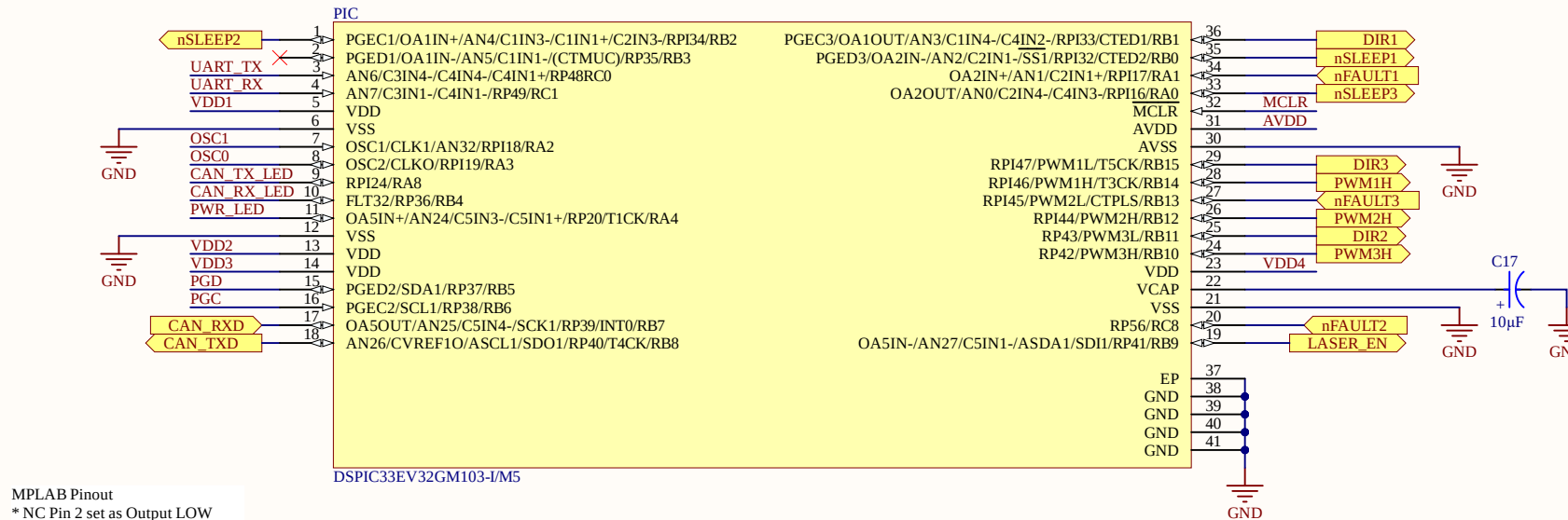
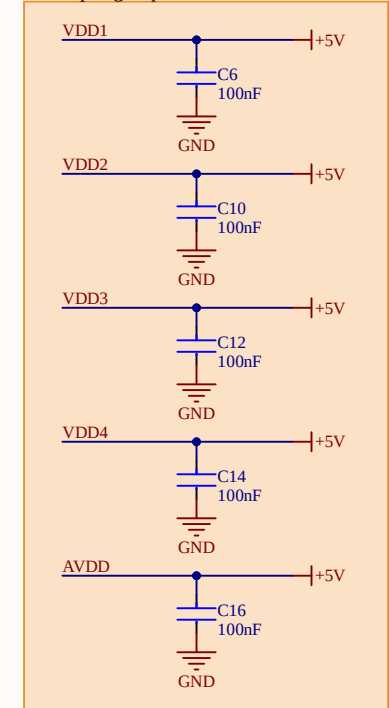
UART Pinheader



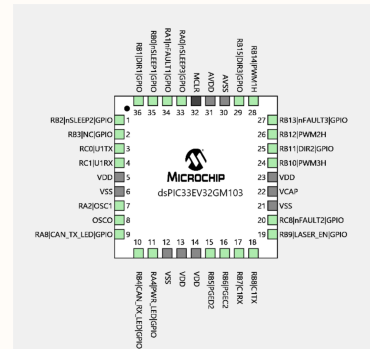
Debug LEDs



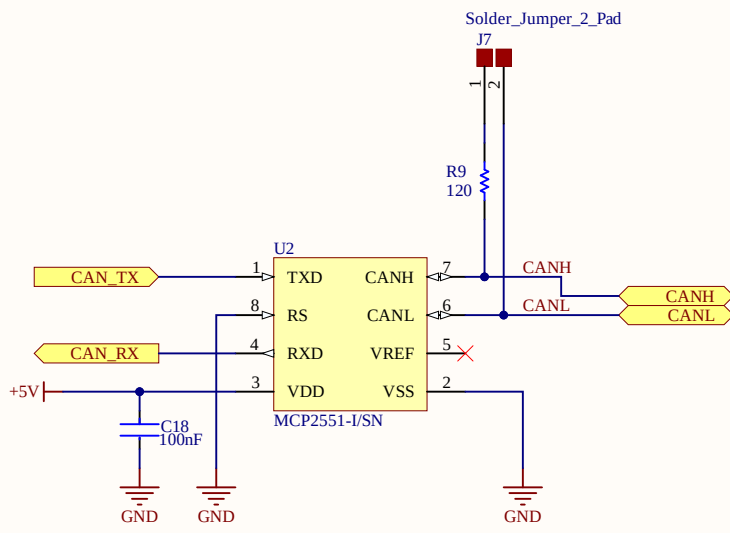
Decoupling Capacitors



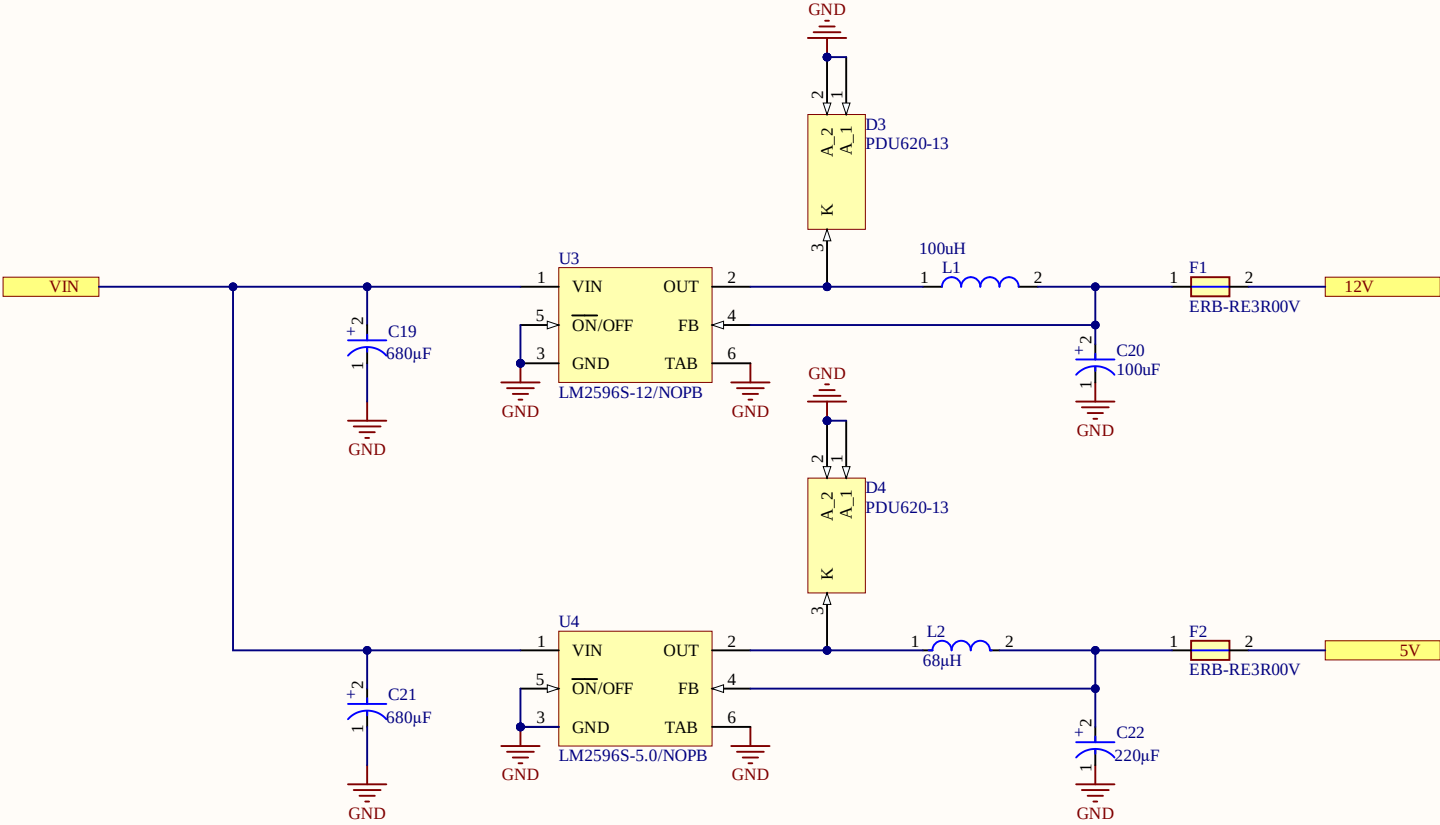
MPLAB Pinout
* NC Pin 2 set as Output LOW



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File: CAN_Transceiver.SchDoc	Drawn By:	

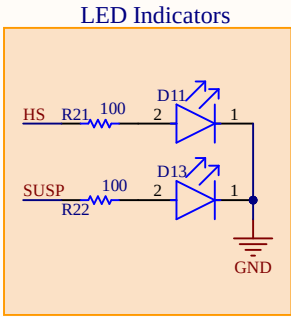
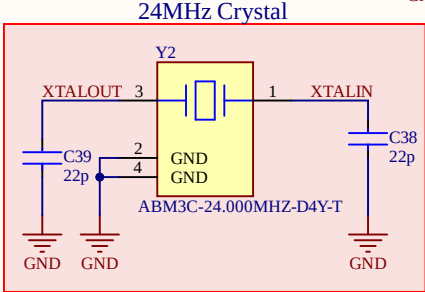
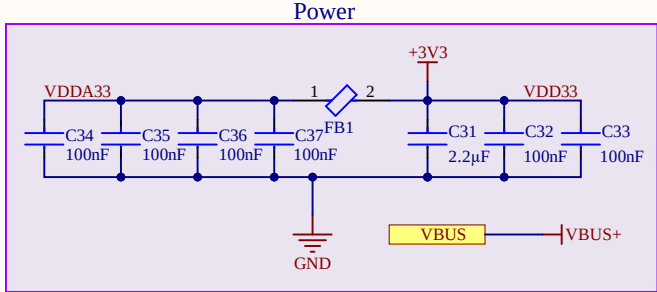
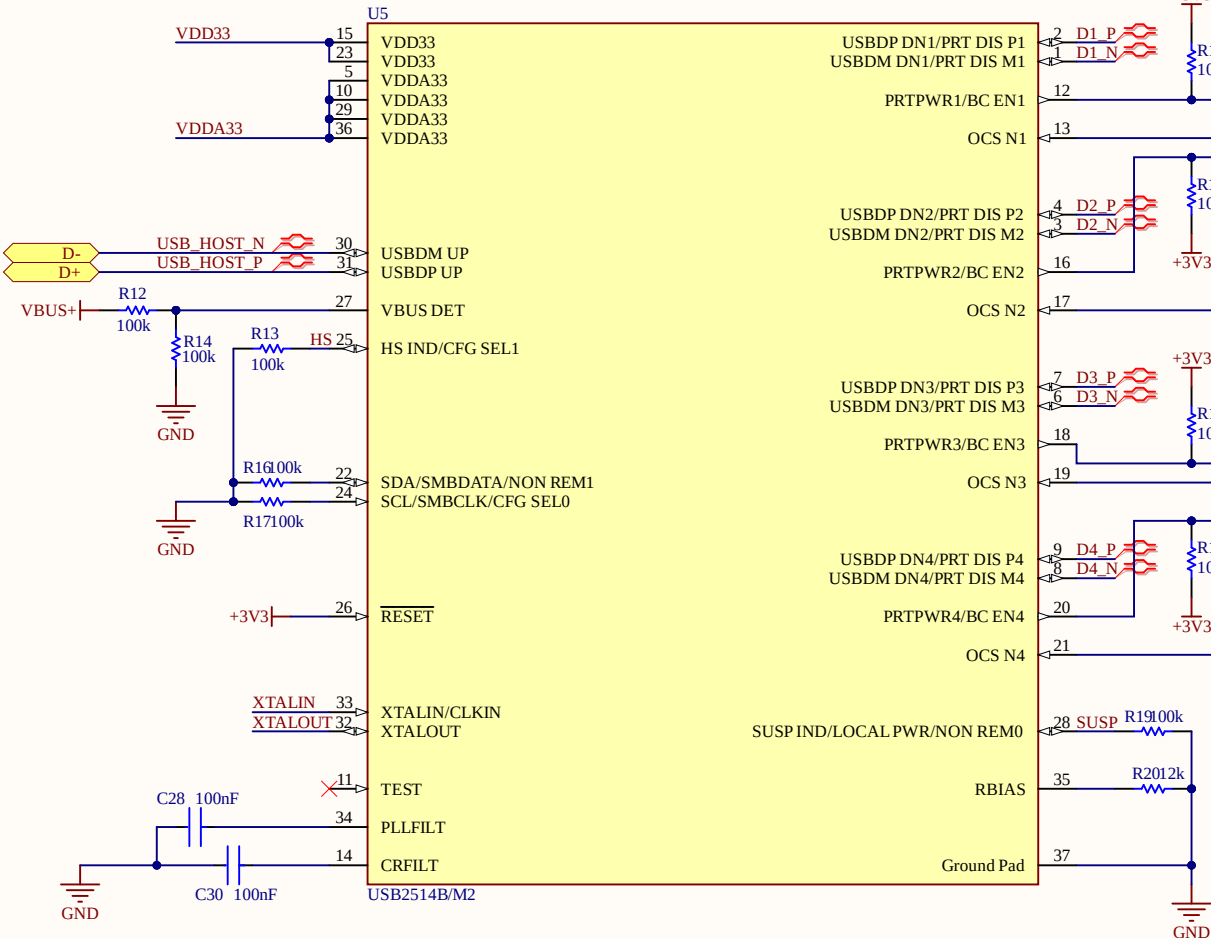


Credits: design modified from the
URC 2022 Science Distribution
Board

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USB Hub Controller

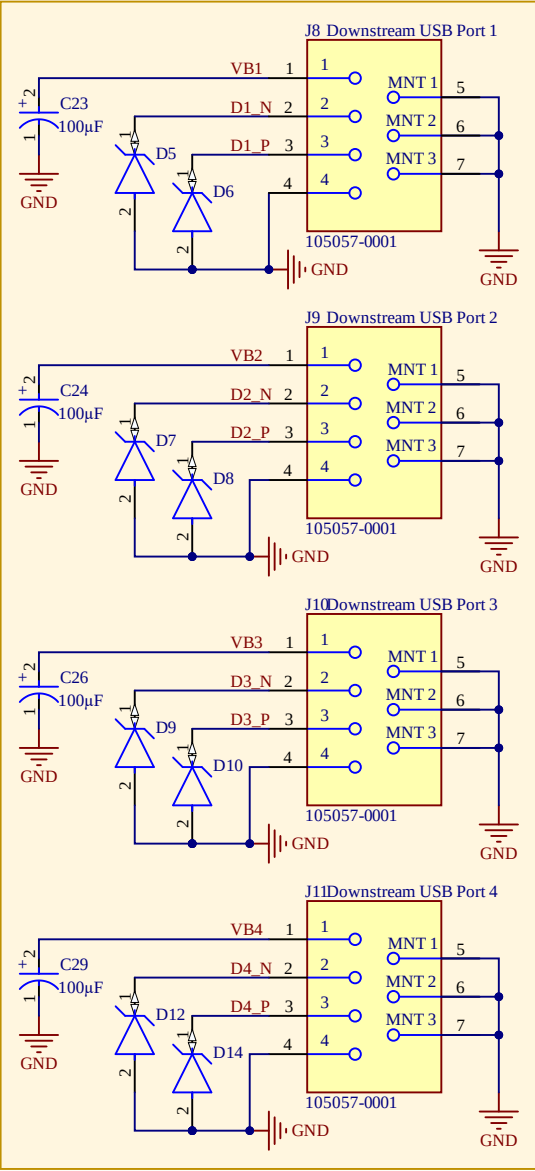
CFG_SEL[1:0] = 0 -> Straps enabled, self-powered, individual power switching & OC sensing
NON-REM[1:0] = 0 -> All ports are removable
OCS_N* pins contain internal pull-ups



Indicator output pins
min 2.4V and max
5.5V @ 12mA

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Downstream Ports w/ ESD Protection

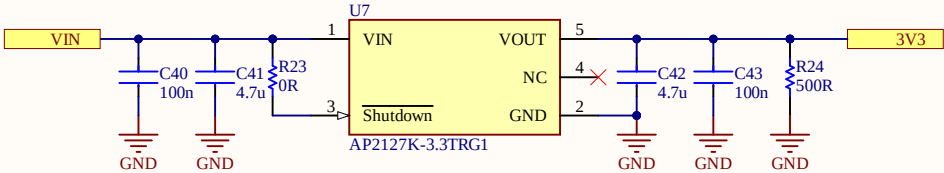


Only USB Hub takes 3.3V
~

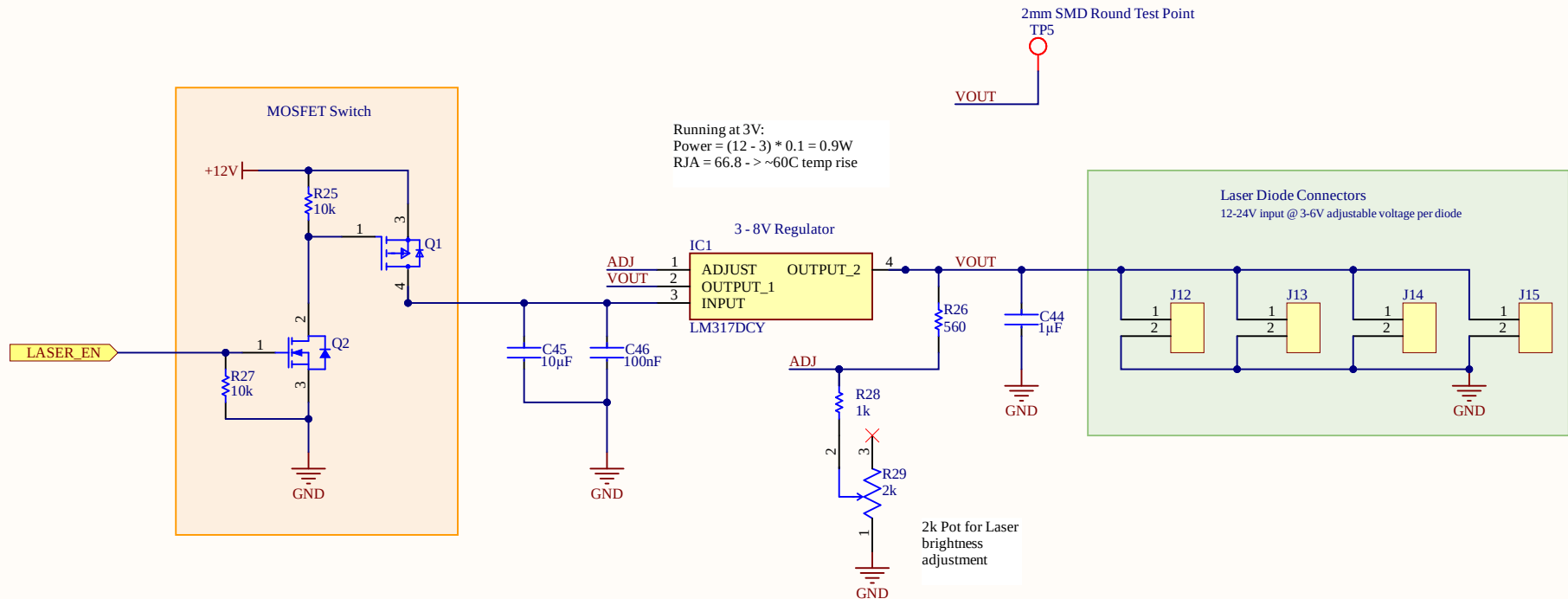
Power = (Vi-Vo) * Io
Power = (5-3.3) * 0.1 = 0.17W

TempRise = Power*theta_JA = 0.17*250 = 42.5 deg Celcius

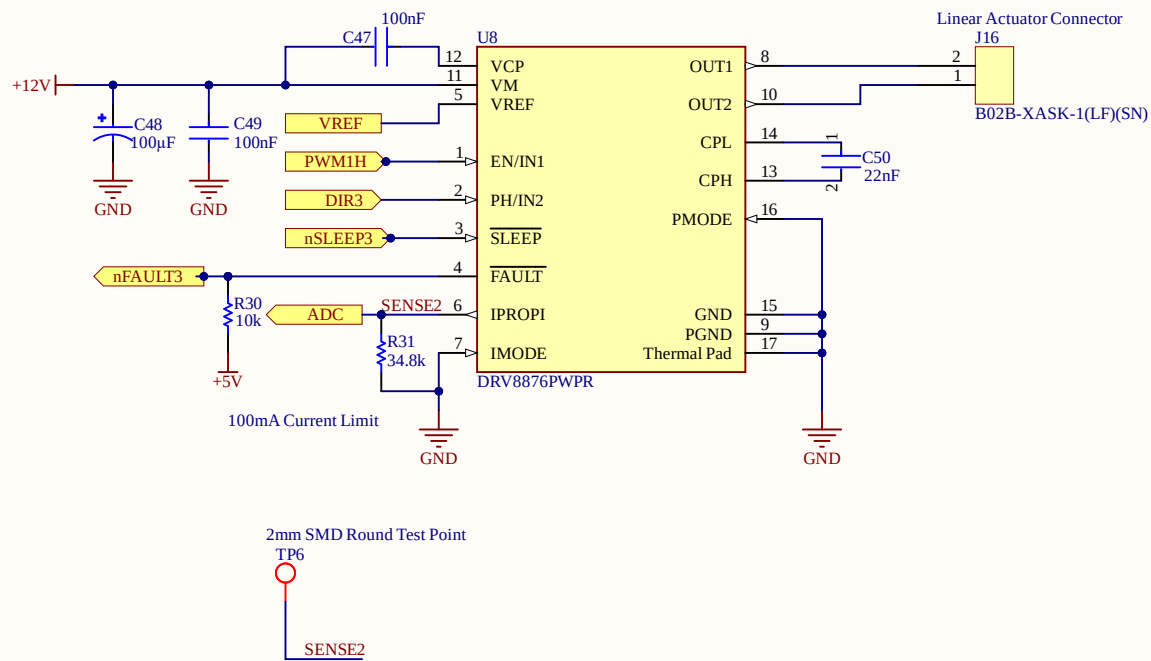
NOTE: We are not too sure on the USB Hub current draw,
so our Io value might be different.
please help :(



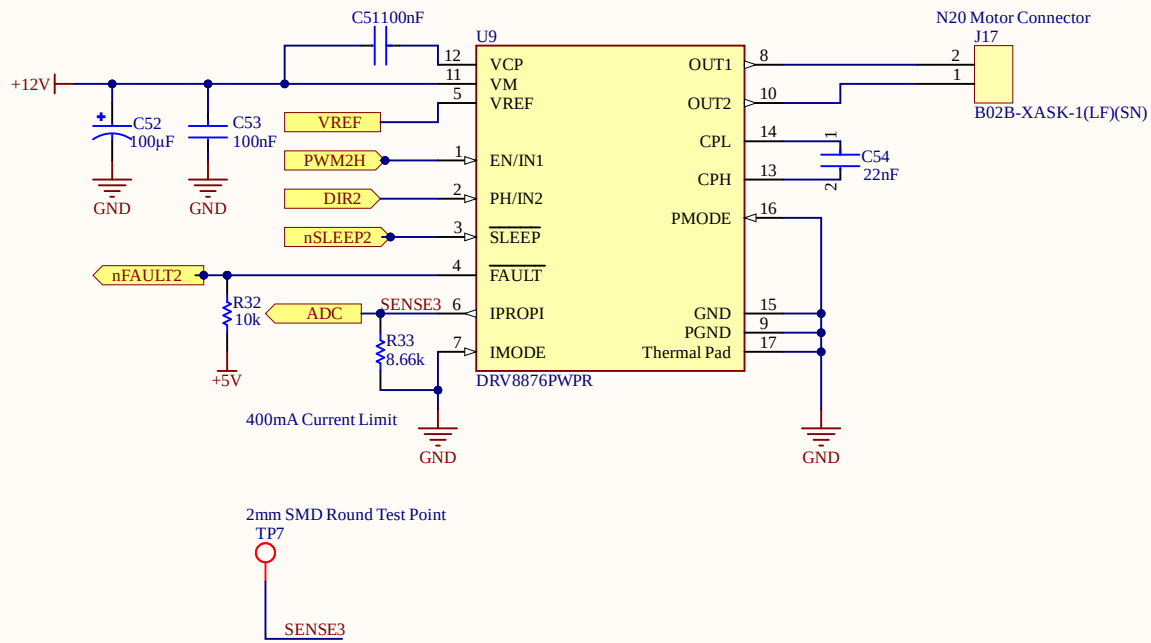
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Title		
Size	Number	Revision
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File:	Laser_Diode_Driver.SchDoc	Drawn By:



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File:	Linear_Actuator_Driver.SchDoc	Drawn By:



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File:	N20_Motor_Driver.SchDoc	Drawn By: